Specialised cells · complex sentences · roots ⏱ ~25 min

NAME _____ DUE _____ SCORE / EFFORT _____

1 · Read

The Right Cell for the Job

¶1 Although every cell in your body began from a single fertilised egg, your cells are not all the same. As an embryo develops, cells become **specialised**: their shape and contents change so that they can carry out one particular job very well. Scientists call this process *differentiation*.

¶2 Red blood cells are a striking example. They are shaped like discs that are pressed in on both sides, which gives them a large surface area for absorbing oxygen. Mature red blood cells in humans have no nucleus, so there is more room for haemoglobin, the iron-containing protein that carries oxygen. Because they lack a nucleus, however, they cannot repair themselves, and each one survives only about four months.

¶3 Nerve cells, or neurons, carry electrical signals around the body. Some have extremely long, thin extensions; the nerve cells that run from the base of the spine to the toes can be around a metre long. Their many branching ends allow a single neuron to connect with thousands of others.

¶4 Plants have specialised cells too. Root hair cells have long, finger-like extensions that push between soil particles, increasing the area through which water can be absorbed. On the underside of leaves, pairs of curved **guard cells** open and close tiny pores called *stomata*, controlling how much water vapour escapes and how much carbon dioxide enters.

¶5 In every case, the structure of the cell matches its function. Biologists use this principle constantly: when they discover a cell with an unusual shape, their first question is usually, “What job does that shape make possible?”

2 · Questions

1. In your own words, what is *differentiation*?

2. Complete the table.

Cell	Special feature	How the feature helps it do its job
Red blood cell		
Neuron		
Root hair cell		
Guard cell		

3. Explain the **trade-off** (advantage and disadvantage) of red blood cells having no nucleus.

4. Find **two** complex sentences in the text. Copy them, underline the main clause and bracket the subordinate clause(s).

a) _____

b) _____

5. Summarise the whole text in **one sentence of no more than 25 words**.

3 · Roots

6. *Haem-* / *hem-* is Greek for "blood"; *neuro-* is Greek for "nerve"; *stoma* is Greek for "mouth".
Use these and the roots from class to define: *haematology*, *neurology*, *stomata*, *erythrocyte* (*erythro-* = red).

4 · At home

7. Teach someone at home the three ideas of cell theory **without looking at your notes**. Then write down one question they asked and your answer.